Tangent & Normal

Mind map

Tangent & normal at a point

- $y - y_1 = m(x - x_1)$
- Normal slope $-\frac{1}{m}$

Parametric / piecewise

- $m = \frac{dy/dt}{dx/dt}$
- Junction: $f'(a^-) = f'(a^+)$

Objective concepts

- Even f: $f'(0) = 0$
- At O: lowest terms $= 0$

Vertical tangent

- $f'(x_0^{\pm})$ same infinity
- $x^{1/3}$ yes, $x^{2/3}$ no

Parametric points

- $(at^2, 2at)$, $\left(ct, \frac{c}{t}\right)$
- $(t, f(t))$ on $y = f(x)$

Ultimate concept

- $f'(t) = \frac{f(t) - y_1}{t - x_1}$
- Common tangent: $f'(t_1) = g'(t_2) = m_{PQ}$

Minimum distance

- Along common normal
- Line: tangent $\parallel$ line
- Circle: $CP - r$

Angle of intersection

- $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$
- Each common point separately

Orthogonal curves

- $m_1 m_2 = -1$
- Eliminate using both equations

Touching curves

- $f = g$, $f' = g'$
- $e^x = kx$: $k = e$; $\ln x = kx$: $k = \frac{1}{e}$

Important lengths

- Subtangent $\left|\frac{y}{m}\right|$, subnormal $|ym|$
- Tangent $\left|\frac{y\sqrt{1+m^2}}{m}\right|$, normal $|y\sqrt{1+m^2}|$

Approximation

- $f(x + \Delta x) \approx f + f'\Delta x$
- $f'' > 0$: under-estimate

Rate measure

- $\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$
- Relation first, values last